

系统设计指南

作者：陈七

私人图书馆 · Private Library Collection

Chapter 1: Scalability Fundamentals

Key properties of reliable systems:

- Availability: system remains operational
- Scalability: handles growing load
- Reliability: works correctly over time
- Maintainability: easy to operate and evolve

Scaling strategies:

- Vertical: upgrade single machine (CPU, RAM)
- Horizontal: add more machines (preferred for large scale)

Load balancers distribute traffic across servers.

Chapter 2: Database Design

Relational databases (PostgreSQL, MySQL):

- ACID transactions
- Strong consistency
- Complex queries with SQL

NoSQL databases:

- Document (MongoDB): flexible schemas
- Key-Value (Redis): ultra-fast caching
- Column (Cassandra): write-heavy workloads
- Graph (Neo4j): relationship queries

Sharding partitions data across nodes; Replication copies data for redundancy.

Chapter 3: Caching and CDN

Caching reduces latency and database load.

Cache patterns:

- Cache-Aside: app checks cache, falls back to DB
- Write-Through: write to cache and DB simultaneously
- Write-Behind: write to cache, async flush to DB

CDN (Content Delivery Network) serves static assets from geographically distributed servers, reducing latency for global users.

Redis supports: strings, hashes, lists, sets, sorted sets, pub/sub.